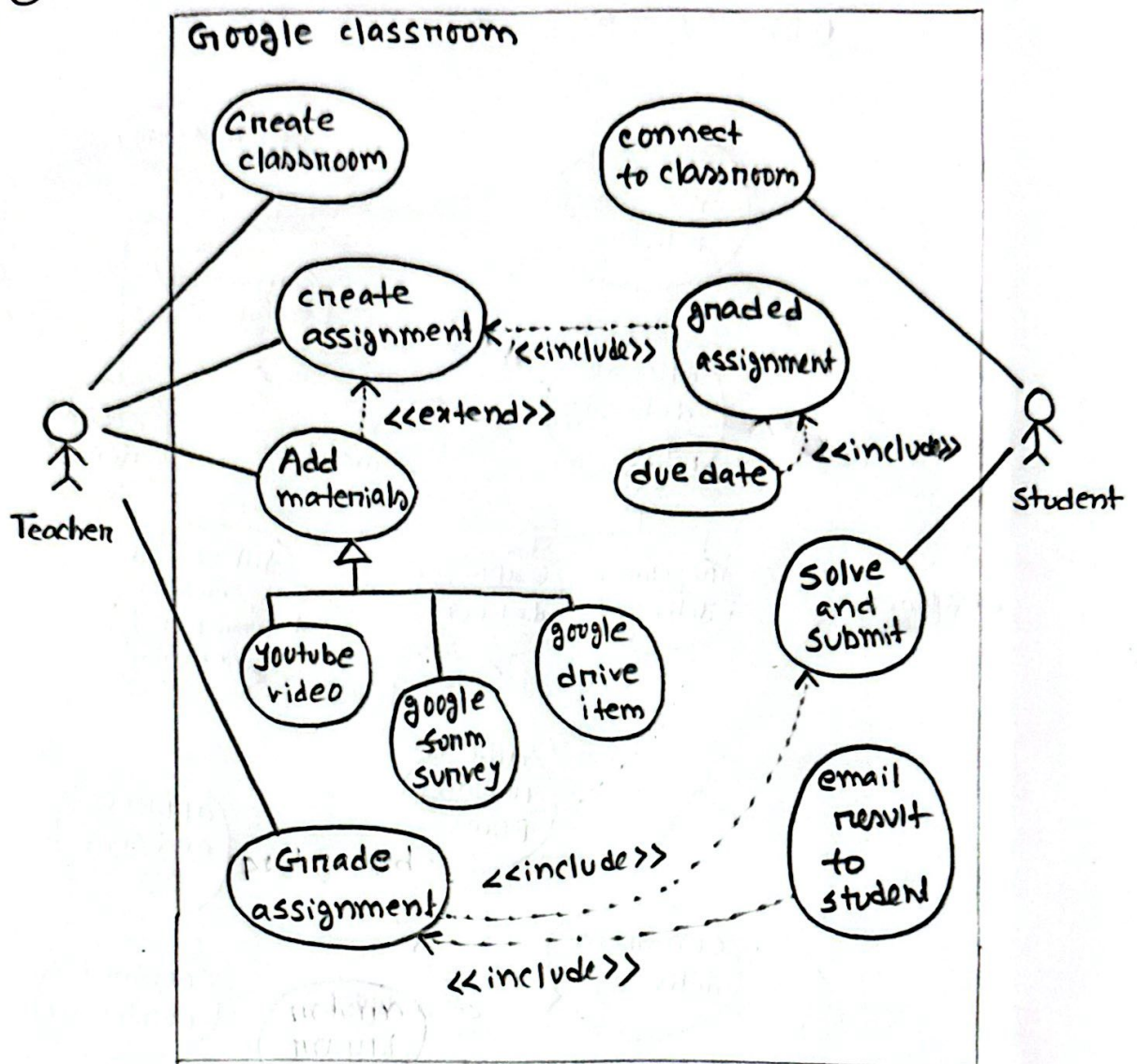


1. Google Classroom is a teacher/student platform, which Google designed for the way teachers and students think and work. It converts traditional paper-based teaching mechanisms to an online paperless one. It all starts with the creation of a classroom by a teacher. Students can easily connect to a classroom using a unique code created for each classroom. It also allows the teachers to create and add materials to assignments, such as YouTube videos, a Google Forms survey, and other items from Google Drive. In this pandemic situation, it has done more for the teacher-student community. Teachers can set graded assignments. Due date setting facility is also offered for the assignments. Once the assignment is posted, students can solve those and submit them online. On completion of submission, teachers can grade and return the result to the students. The updates of the events are also emailed to corresponding email addresses which are used while connecting to the classroom.

Design a use case diagram based on the above information.

Use case diagram

①

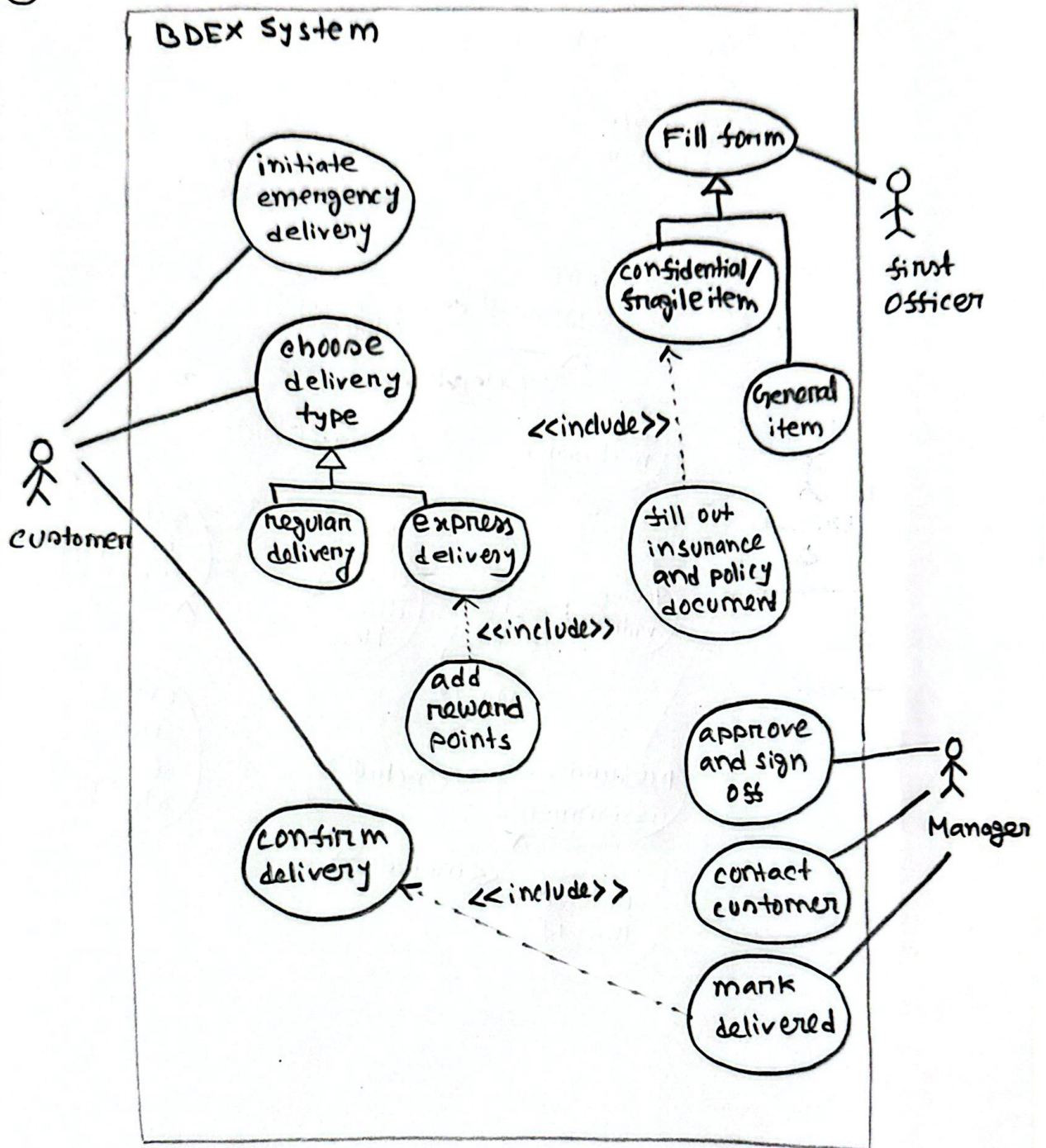


2. Analyzing the system for BDEX (a Bangladeshi courier service), following requirements were identified:

1. There will be three types of users: Customer, First Officer and Manager
2. Customer can initiate emergency delivery by providing item details
3. Customer can choose between regular and express delivery
4. Reward points are added into Customer account for express delivery
5. First Officer can fill out courier form with necessary information
6. For confidential and/or fragile item(s), First Officer must fill out insurance and policy documents during packaging but it is not needed for general items.
7. Manager can approve and sign off parcel item(s) for delivery
8. Manager can contact Customer for delay in parcel arrival
9. Manager can mark packages delivered after getting Customer confirmation.

Design a use case diagram based on the above information.

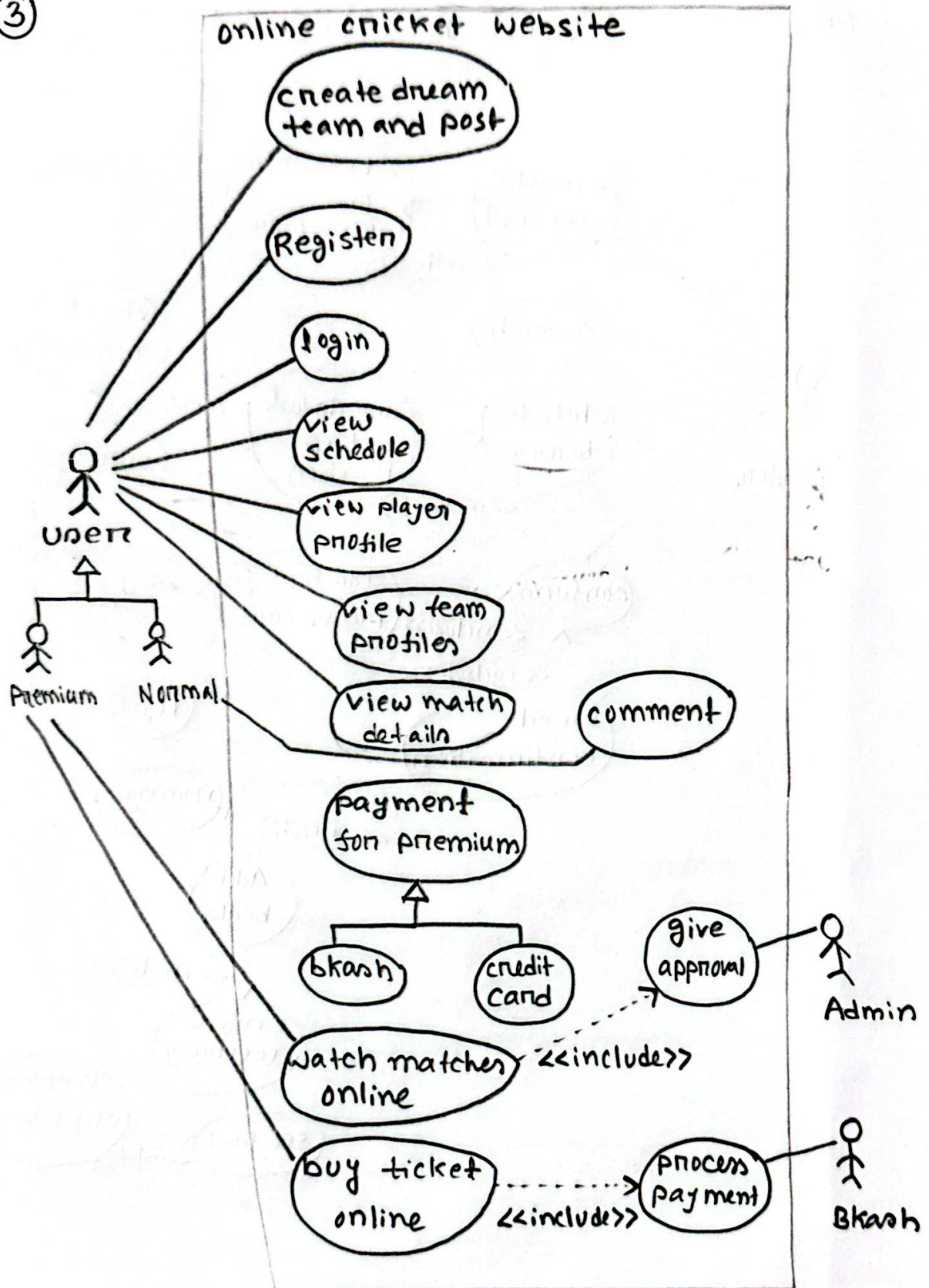
2



3) BDXICRIC is planning to create an online website which consists of detailed information of every match in BPL. It is specially made for cricket lovers. In this online website users have to register for free access. After login to the website, the user can view schedule, player profile, Team profiles, match details. Users can update his/her status to Premium by paying a fee. Payment can be done by BKASH or Credit card. Premium users can watch matches online but special approval is needed from the administrator. Premium users can buy tickets online which are handled by BKASH. All users can create a dream team and post it on the website. Other users can comment in the post.

Design a use case diagram based on the above scenario.

③

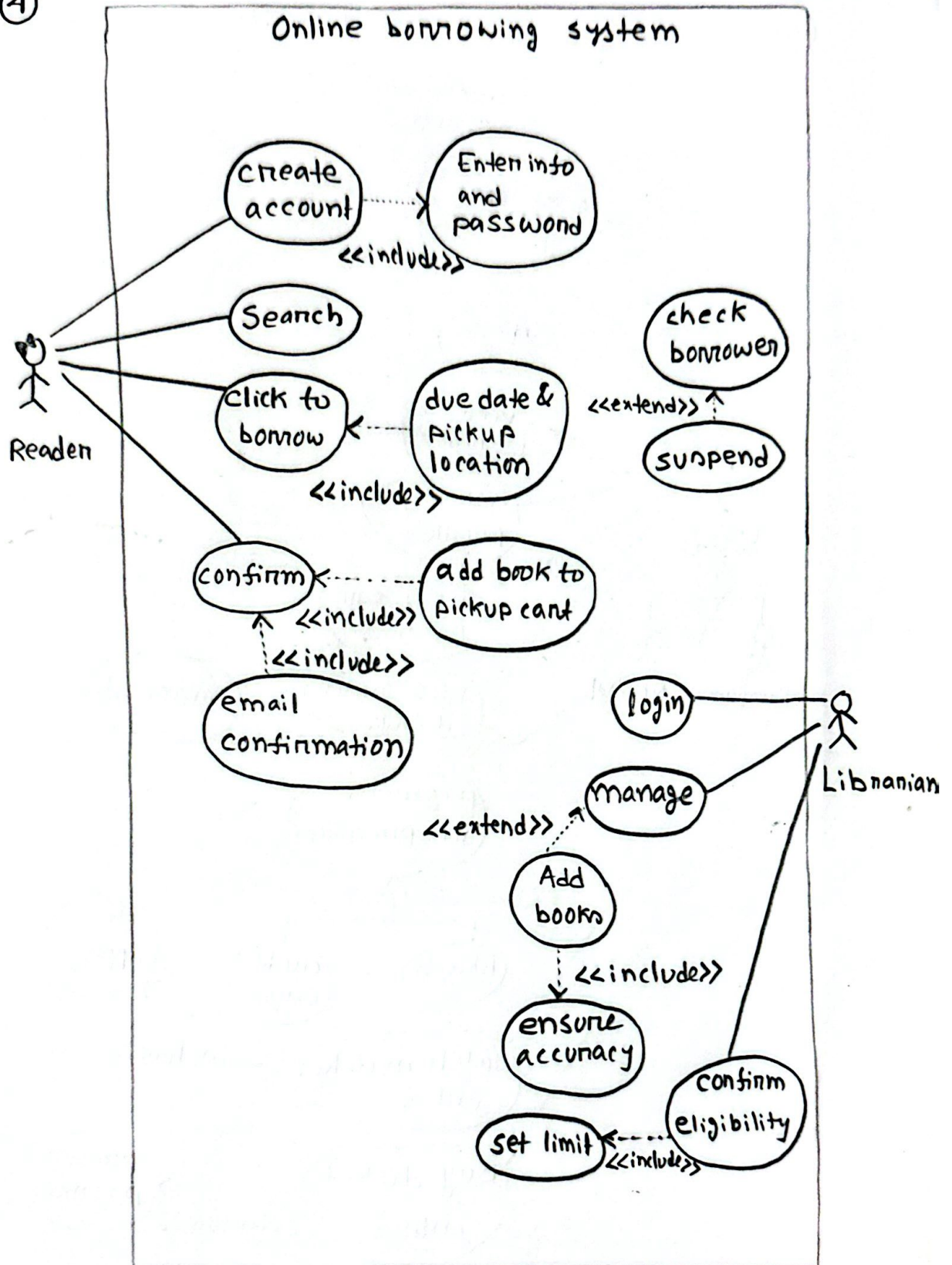


4) Design a USE CASE diagram from the below scenario.

Sarah, an avid reader, decides to use the library's new online borrowing system. She first creates an account, entering her information and setting a password. Excited to explore the vast collection, she searches for "The Martian" by Andy Weir . Finding the book available, Sarah clicks to borrow it, and a window pops up displaying the due date and pickup location. Satisfied, she confirms the borrow which adds the book to the pick-up cart. When she confirms the book in the cart window, an email confirmation arrives, informing her of the pickup details.

Meanwhile, at the library, Mr. Lee, the head librarian, logs into the system to manage the collection . He adds several new science fiction novels to the database, including "The Martian," and ensures their information is accurate. Mr. Lee also checks Sarah's account to confirm her borrowing eligibility and sets a limit on the number of books she can borrow at once. He also notices a user with overdue books and decides to temporarily suspend their borrowing privileges.

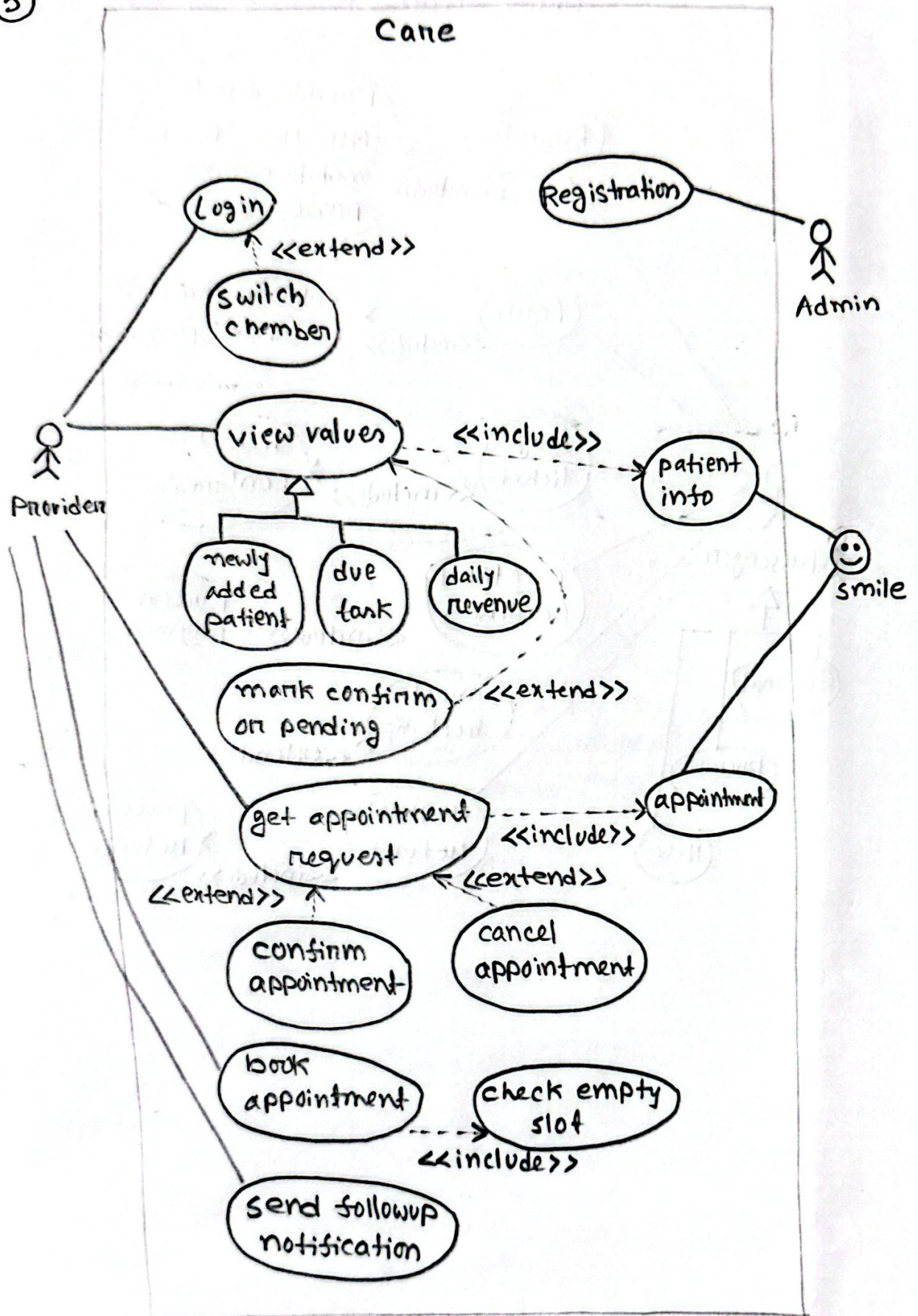
④



- 5) **Care** is a Dental Practice Management system. Several Dental Providers take services from **Care**. Providers can log in to the system but the registration has to be done by the Admin. After login, Providers can switch between chambers if there are multiple chambers of a provider. There is this external system named **Smile** from where the system gets all the patient information. From that information, providers can view 3 values, those are the number of newly added patients, due tasks, and daily revenue. For each task, the provider can mark it as Confirm or Pending. The provider can get information about patient appointment requests from the external system. And thus confirm or cancel the appointment. A provider can also book appointments for the patient directly from the system in a scheduler. But for this provider has to check the scheduler for an empty slot. Lastly, the provider can send Follow up notifications to the patients.

Now, you have to draw a **Use Case Diagram** based on the following scenario.

⑤

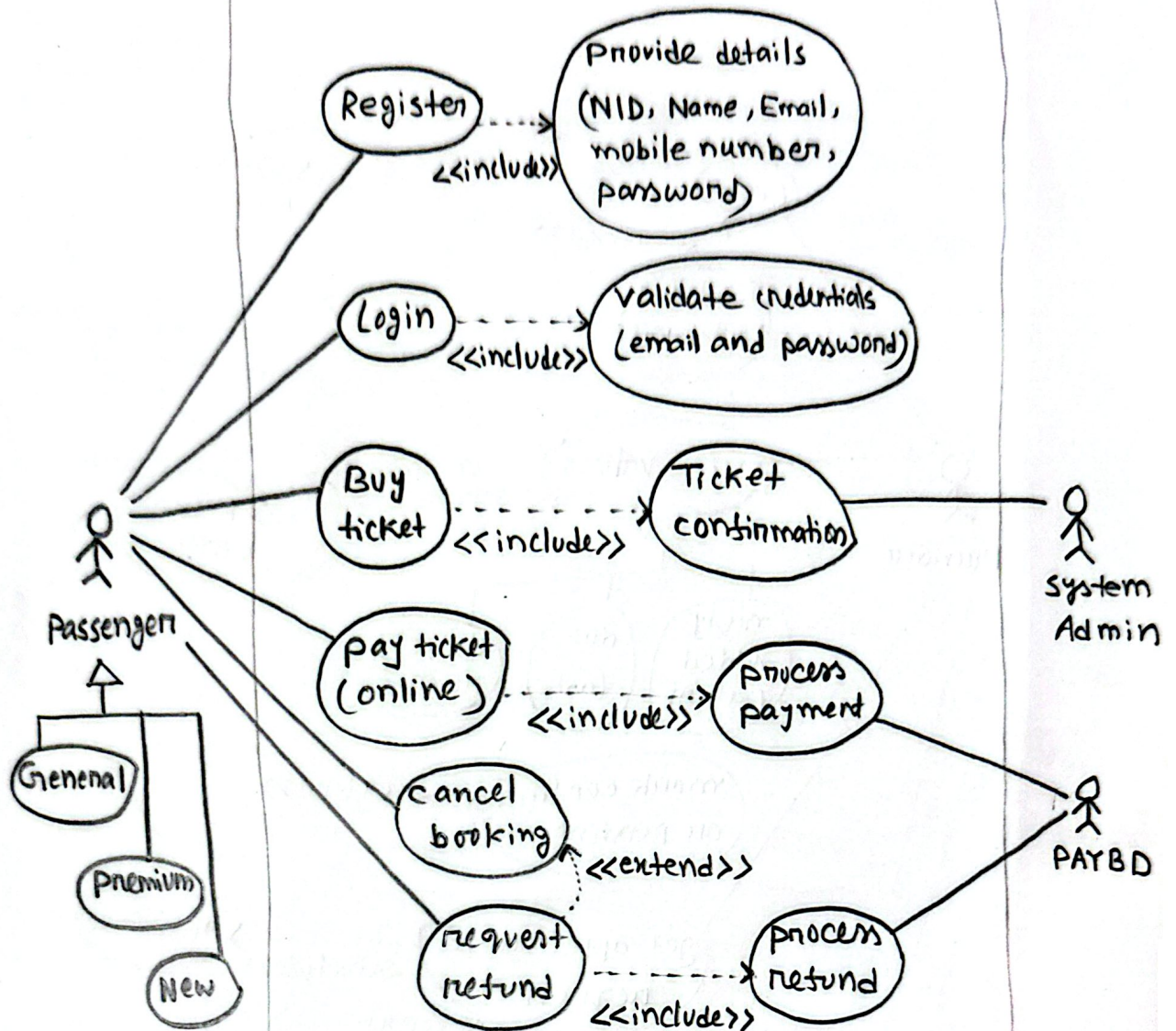


6. Analyzing the system for online ticket booking, the following requirements were identified

1. There can be three types passenger: General, Premium and New
2. Passenger can register into the system with his – NID, Name, Email, Mobile Number, and Password
3. Passenger can login into the system with his email and password
4. Passenger can buy ticket
5. Ticket should be confirmed by the System Admin
6. Passenger can pay the ticket price online after being confirmed
7. Online payment should be using the “PAYBD” payment gateway
8. Passenger can cancel a booking
9. Passenger can request for return money for the canceled booking

Design a Use Case diagram based on the above information.

Online Ticket Booking System



7. Analyzing the system for student assignment management, the following requirements were identified

1. There can be three types of users: Teacher, Teaching Assistant (TA) and

Student 2. Teacher can add new assignments

3. Teacher can edit/remove an existing assignment

4. TA can edit/remove an existing assignment

5. Student can submit an assignment

6. Student can edit his submission (before due date)

7. TA can assign grade to submission (after due date)

8. Teacher can modify grade after graded by the TA

9. Student can view grade

10. Student can generate reports of his submissions after graded by the

TA Design a Use Case diagram based on the above information.

